

Topic 1: Introduction – Distribution function

- In statistics and data science, understanding how data values are spread is very important. The distribution function explains how the values of a random variable are distributed over a particular range. In simple words, it tells us the pattern or behavior of data.

Continuous Function

Before learning about the distribution function, we must understand a continuous function.

A continuous function is a function that does not have sudden jumps, gaps, or breaks. The values change smoothly from one point to another.

Example:

If we draw the graph of this function, it forms a smooth curve. This smooth nature is important when dealing with continuous probability distributions.

Probability Density Function (PDF)

For continuous random variables, the distribution is represented using a Probability Density Function (PDF).

The PDF shows how probability is distributed across different values of a variable. It does not give probability at a single point, but gives probability over an interval.

Conditions of PDF

1. The function should not have negative values
2. The total area under the curve must be equal to 1

The probability of a value lying between two points is calculated as the area under the curve between those points.

Probability Mass Function (PMF)

When the random variable is discrete, we use Probability Mass Function (PMF) instead of PDF.

PMF directly assigns probability to each possible value of the variable.

Cumulative Distribution Function (CDF)

The Cumulative Distribution Function (CDF) gives the probability that a random variable is less than or equal to a specific value.

For continuous variables, the CDF represents the total accumulated area under the PDF curve from negative infinity to x .

Types of Probability Distributions

Some common probability distributions are:

- Uniform Distribution – all values in a range have equal probability
- Normal Distribution – symmetric bell-shaped distribution
- Exponential Distribution – models time between events
- Binomial Distribution – used for discrete success/failure experiments

Each distribution is used for different types of data and different real-world situations.

Topic 2: Uniform distribution

- Uniform distribution is one of the simplest probability distributions. In this distribution, all values within a given range have equal probability.
- In simple words, every value between a minimum value a and maximum value b has the same chance of occurring. There is no bias toward any particular value.

Mathematical Representation

For a continuous uniform distribution, the Probability Density Function (PDF) is defined as:

Where:

- a = starting value
- b = ending value

This means the height of the distribution curve is constant between a and b .

- All values in the interval are equally likely.
- The graph of uniform distribution is a rectangle.
- The total area under the curve is equal to 1.
- Mean of uniform distribution:

- Variance:

In uniform distribution:

- The probability does not increase or decrease.
- The curve remains flat between a and b.
- Outside the interval, probability is 0.
- This type of distribution is useful when we assume no preference or randomness is equally spread.

Code:

```
# Import required libraries
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from scipy.stats import uniform
```

```
# Generate uniform data
```

```
data = uniform.rvs(loc=10, scale=20, size=10000)
```

```
# Plot histogram
```

```
sns.histplot(data, bins=50, kde=True)
```

```
plt.title("Uniform Distribution")
```

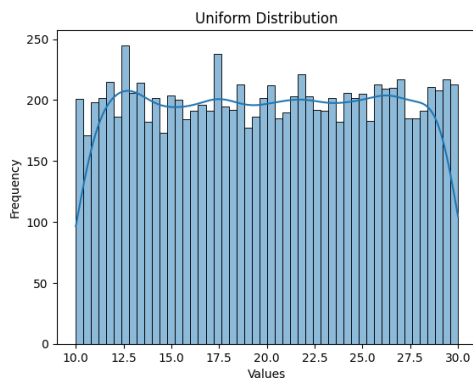
```
plt.xlabel("Values")
```

```
plt.ylabel("Frequency")
```

```
plt.show()
```

- `loc=10` Starting value (a)
- `scale=20` Width (b - a)
- `size=10000` Number of random values generated
- `histplot()` Used to display the distribution graph

Output:



Topic 3: Normal distribution

- The Normal Distribution, also called the Gaussian Distribution, is one of the most important probability distributions in statistics and data science.
- It is a continuous probability distribution that forms a bell-shaped curve. This distribution is symmetric, which means the left side and right side of the graph are mirror images.

In normal distribution:

- Most of the values are concentrated around the mean.
- Fewer values are found as we move away from the mean.
- The curve is perfectly symmetrical.

It is mathematically represented as:

Where:

- = Mean (average)
- = Standard deviation
- = Variance

1. The curve is bell-shaped.
2. Mean = Median = Mode (all are equal).

3. The total area under the curve is equal to 1.
4. The spread of the curve depends on the standard deviation.
5. If standard deviation is small curve is narrow.

 If standard deviation is large curve is wide.

Normal distribution is important because:

- Many real-world data follow this distribution (height, marks, weight, etc.).
- It is based on the Central Limit Theorem.
- It is widely used in statistical analysis and machine learning.

Code:

```
# Import required libraries
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from scipy.stats import norm
```

```
# Generate normal distributed data
```

```
data = norm.rvs(loc=50, scale=10, size=1000)
```

```
# Plot histogram
```

```
plt.hist(data, bins=30, density=True)
```

```
# Create x values
```

```
x = np.linspace(10, 90, 100)
```

```
# Plot normal distribution curve
```

```
plt.plot(x, norm.pdf(x, 50, 10))
```

```
plt.title("Normal Distribution")
```

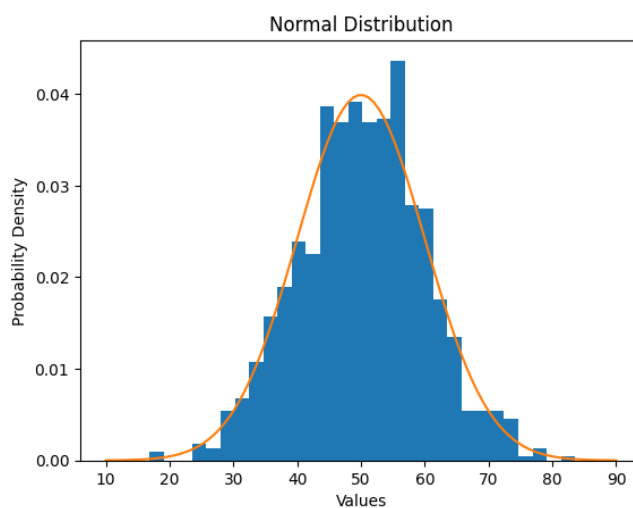
```
plt.xlabel("Values")
```

```
plt.ylabel("Probability Density")
```

```
plt.show()
```

- `loc=50` Mean value is 50
- `scale=10` Standard deviation is 10
- `size=1000` Number of random values generated
- `norm.pdf()` Used to plot probability density curve

Output:



Topic 4: Exponential Distribution

- The Exponential Distribution is a continuous probability distribution. It is mainly used to describe the time between events that happen continuously and independently at a constant average rate.
- In simple words, it tells us how long we need to wait until the next event happens.

For example:

- Time between customer arrivals in a bank
- Time between system failures
- Time between incoming network requests

Probability Density Function (PDF)

The probability density function of exponential distribution is:

Where:

- = rate parameter
- = exponential constant
-

Important points:

- The function is defined only for positive values.
- The graph decreases continuously as x increases.
- The total area under the curve is equal to 1.
- This means that smaller waiting times have higher probability, and as time increases, probability decreases.

Characteristics of Exponential Distribution

- It is a continuous distribution.
- It is positively skewed (tail on right side).
- It models time-based events.
- It is memoryless (previous events do not affect future events).

Code:

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from scipy.stats import expon
```

```
# Generate random data
```

```
data = expon.rvs(scale=1, loc=0, size=1000)
```

```
# Plot histogram
```

```
sns.histplot(data, bins=50, kde=True)
```

```
plt.xlabel("Exponential Distribution")
```

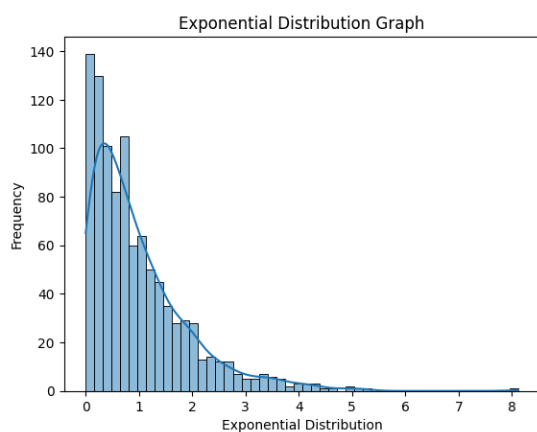
```
plt.ylabel("Frequency")
```

```
plt.title("Exponential Distribution Graph")
```

```
plt.show()
```

- `expon.rvs()` generates random values from exponential distribution
- `scale` controls spread of distribution
- `size` number of random values generated
- `histplot()` draws histogram
- `kde=True` draws smooth curve

Output:



Topic 5: Binomial Distribution

- The Binomial Distribution is a type of probability distribution used for discrete random variables. It is applied when an experiment has only two possible outcomes, such as:

✓ Success / Failure

✓ Yes / No

✓ True / False

✓ Head / Tail

- It is mainly used when we repeat an experiment several times under the same conditions.

In binomial distribution:

- Each trial has only two outcomes.
- The probability of success remains constant.
- Each trial is independent of the others.

If we perform an experiment n times, and the probability of success in each trial is p , then the binomial distribution helps us calculate the probability of getting a certain number of successes.

Formula

The probability formula of binomial distribution is:

Where:

- = number of trials
- = number of successes
- = probability of success
- = probability of failure

This formula gives the probability of getting exactly k successes in n trials.

Example

Suppose:

- We toss a coin 10 times
- Probability of getting head (success) = 0.5
- We want to know probability of getting 6 heads

This situation follows binomial distribution because:

- Only two outcomes (Head or Tail)
- Same probability each time

- Each toss is independent

Graph Nature

The binomial distribution graph:

- Is discrete (separate bars, not continuous curve)
- Can be symmetric if $p = 0.5$
- Can be skewed if p is closer to 0 or 1

Code:

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from scipy.stats import binom
```

```
# Number of trials
```

```
n = 10
```

```
# Probability of success
```

```
p = 0.8
```

```
# Generate random data
```

```
data = binom.rvs(n=n, p=p, size=10000)
```

```
# Plot histogram
```

```
plt.hist(data, bins=10)
```

```
plt.xlabel("Number of Successes")
```

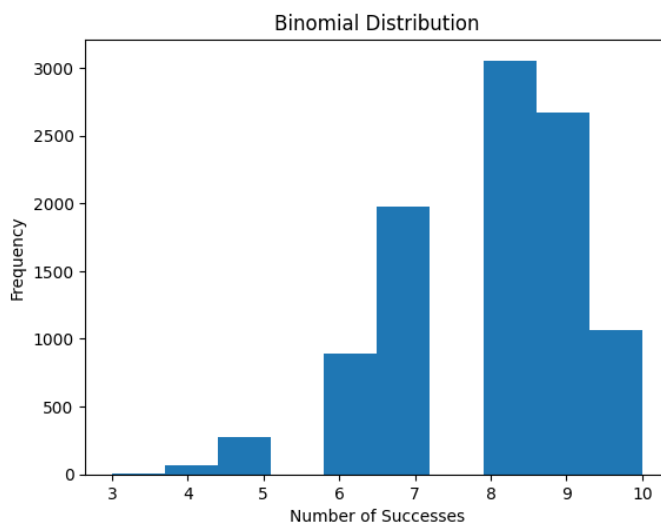
```
plt.ylabel("Frequency")
```

```
plt.title("Binomial Distribution")
```

```
plt.show()
```

- $n = 10$ total number of trials
- $p = 0.8$ probability of success
- $\text{size} = 10000$ number of random samples generated
- `plt.hist()` draws the graph

Output:



Binomial distribution is used in:

- Quality testing (defective / non-defective items)
- Survey analysis (Yes/No responses)
- Medical testing (positive / negative result)
- Machine learning classification problems

Topic 6: Cumulative Distribution function

- The Cumulative Distribution Function (CDF) is used to find the probability that a random variable is less than or equal to a particular value.
- In simple words, CDF tells us how much probability has accumulated up to a

certain point.

- Mathematically, it is written as:
-
- This means the probability that the random variable takes a value less than or equal to .
- If we already have a Probability Density Function (PDF), then the CDF is obtained by finding the area under the PDF curve from negative infinity to x .

So we can say:

- PDF gives density of probability
- CDF gives accumulated probability

For continuous data, the CDF curve always:

- Starts from 0
- Ends at 1
- Never decreases (it is always increasing)

Important Properties of CDF

- 1.
2. It is a non-decreasing function
3. As ,
4. As ,

Code:

```
# Cumulative Distribution Function (CDF) Example
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from scipy.stats import norm
```

```
# Generate normal distribution data

data = norm.rvs(loc=0, scale=1, size=1000)


# Sort the data

data_sorted = np.sort(data)


# Calculate CDF values

cdf = norm.cdf(data_sorted, loc=0, scale=1)


# Plot the CDF

plt.plot(data_sorted, cdf)

plt.title("Cumulative Distribution Function (CDF)")

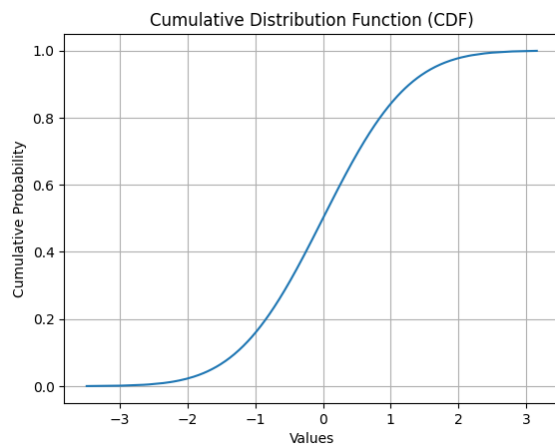
plt.xlabel("Values")

plt.ylabel("Cumulative Probability")

plt.grid(True)

plt.show()
```

Output:



Topic 7: Measures of Central Tendency

- In statistics, Measures of Central Tendency are used to find the central or average value of a dataset.
- In simple words, they help us understand the typical value around which the data is distributed.
- When we analyze data, instead of looking at all values, we summarize the dataset using one representative value. That representative value is called the measure of central tendency.

The three main measures are:

- ✓ Mean
- ✓ Median
- ✓ Mode

Mean (Average)

The Mean is the sum of all values divided by the total number of values.

Mathematically,

Where:

- = Sum of all observations
- = Number of observations

Example

If the dataset is:

The mean is affected by extreme values (outliers).

Median

The Median is the middle value of the dataset when the data is arranged in ascending or descending order.

- If the number of observations is odd Middle value
- If the number of observations is even Average of two middle values
- Median is not affected much by extreme values.

Mode

The Mode is the value that appears most frequently in the dataset.

Example:

Mode = 3 (because it appears twice)

A dataset can have:

- One mode (Unimodal)
- Two modes (Bimodal)
- More than two modes (Multimodal)

Code:

```
import numpy as np
```

```
import statistics as stats
```

```
# Sample data
```

```
data = [2, 4, 6, 8, 10, 6, 4]
```

```
# Calculate Mean
```

```
mean_value = np.mean(data)
```

```
# Calculate Median
```

```
median_value = np.median(data)
```

```
# Calculate Mode
```

```
mode_value = stats.mode(data)
```

```
# Print results
```

```
print("Mean:", mean_value)
```

```
print("Median:", median_value)
```

```
print("Mode:", mode_value)
```

Output:

Mean: 5.714285714285714

Median: 6.0

Mode: 4

Topic 8: Measures of Dispersion

- In statistics, after finding the average of the data, we also need to understand how the data values are spread. The Measures of Dispersion tell us how much variation or spread is present in a dataset.
- In simple words, while mean tells us the center of the data, dispersion tells us how far the values are from the center.
- If dispersion is small values are close to the mean.
If dispersion is large values are widely spread.

1. Range

The Range is the difference between the maximum and minimum values in the dataset.

It is the simplest measure of dispersion.

2. Variance

Variance measures how far each value in the dataset is from the mean.

It is calculated as:

Where:

- = each value
- = mean
- = number of values

Variance gives squared units.

3. Standard Deviation

The Standard Deviation is the square root of variance.

It tells us the average distance of each value from the mean.

If standard deviation is:

- Low data is closely grouped
- High data is widely spread

4. Interquartile Range (IQR)

The Interquartile Range (IQR) measures the spread of the middle 50% of the data.

Where:

- Q1 = 25th percentile
- Q3 = 75th percentile

IQR is useful because it is not affected much by outliers.

5. Skewness

Skewness measures the asymmetry of the distribution.

- Positive skew tail on right side
- Negative skew tail on left side
- Zero skew symmetric distribution

6. Kurtosis

Kurtosis measures the heaviness of the tails of a distribution.

- High kurtosis more outliers
- Low kurtosis fewer outliers

Code:

```
import numpy as np
```

```
import pandas as pd
```

```
# Sample data
```

```
data = [10, 12, 15, 18, 20, 22, 25, 30]
```

```
# Convert to pandas Series
```

```
series = pd.Series(data)
```

```
# Range
```

```
data_range = series.max() - series.min()
```

```
# Variance
```

```
variance = series.var()
```

```
# Standard Deviation
```

```
std_dev = series.std()
```

```
# Interquartile Range (IQR)
```

```
Q1 = series.quantile(0.25)
```

```
Q3 = series.quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
# Skewness
```

```
skewness = series.skew()
```

```
# Kurtosis
```

```
kurtosis = series.kurt()
```

```
# Print results
```

```
print("Range:", data_range)
```

```
print("Variance:", variance)
```

```
print("Standard Deviation:", std_dev)
```

```
print("IQR:", IQR)
```

```
print("Skewness:", skewness)
```

```
print("Kurtosis:", kurtosis)
```

Output:

Range: 20

Variance: 44.857142857142854

Standard Deviation: 6.697547525560595

IQR: 8.5

Skewness: 0.2776945746733307

Kurtosis: -0.59963893058542

Topic 9: Calculating Percentiles

- In statistics, percentiles are used to understand the relative position of a value in a dataset. A percentile tells us what percentage of data values lie below a particular value.
- In simple words, a percentile shows how a value compares with the rest of the data.
- For example:

If a student's score is in the 80th percentile, it means the student scored better than 80% of the students.

The percentile of a value can be understood using the formula:

This formula gives the percentage of data that is smaller than a given value.

Consider the dataset:

If we want to find the percentile of value 4:

- Number of values less than 4 = 4
- Total number of values = 12

This means 33.33% of values are less than 4.

Common Percentiles

Some commonly used percentiles are:

- 25th Percentile First Quartile (Q1)
- 50th Percentile Median (Q2)
- 75th Percentile Third Quartile (Q3)

The 50th percentile is equal to the median.

Importance of Percentiles

- Helps in ranking and comparison
- Used in exams and competitive tests
- Helps understand data distribution
- Useful in boxplot and statistical analysis
- Helps in detecting outliers

Code:

```
import numpy as np

# Sample data
data = [10, 12, 15, 18, 20, 22, 25, 30]

# Calculate different percentiles
p25 = np.percentile(data, 25)
p50 = np.percentile(data, 50) # Median
p75 = np.percentile(data, 75)

# Print results
print("25th Percentile (Q1):", p25)
print("50th Percentile (Median):", p50)
print("75th Percentile (Q3):", p75)
```

Output:

25th Percentile (Q1): 14.25

50th Percentile (Median): 19.0

75th Percentile (Q3): 22.75

Topic 10: Quartiles

- In statistics, Quartiles are the values that divide a sorted dataset into four equal parts. Each part represents 25% of the data.
- In simple words, quartiles help us understand how the data is distributed across four sections.
- Before calculating quartiles, the data must be arranged in ascending order.

Types of Quartiles

There are three main quartiles:

1. First Quartile (Q1)

- Also called the 25th percentile
- 25% of the data lies below Q1
- It represents the lower part of the dataset

2. Second Quartile (Q2)

- Also called the 50th percentile
- It is equal to the Median
- 50% of the data lies below Q2

3. Third Quartile (Q3)

- Also called the 75th percentile
- 75% of the data lies below Q3
- It represents the upper portion of the dataset

Importance of Quartiles

- They help in understanding data distribution clearly.
- They divide data into four equal groups.
- They are useful in detecting outliers.
- They are used in box plot representation.

- They help in calculating Interquartile Range (IQR).

Interquartile Range (IQR)

- The Interquartile Range (IQR) measures the spread of the middle 50% of data.
-
- It is not highly affected by extreme values.

Example

Suppose we have the dataset:

10, 15, 20, 25, 30, 35, 40, 45

After sorting (already sorted here):

- Q1 = 25th percentile
- Q2 = Median
- Q3 = 75th percentile

These values divide the data into four equal parts.

Code:

```
import numpy as np
```

```
import pandas as pd
```

```
# Sample data
```

```
data = [10, 15, 20, 25, 30, 35, 40, 45]
```

```
# Convert to pandas Series
```

```
series = pd.Series(data)
```

```
# Calculate Quartiles

Q1 = series.quantile(0.25)

Q2 = series.quantile(0.50) # Median

Q3 = series.quantile(0.75)


# Calculate IQR

IQR = Q3 - Q1


# Print results

print("First Quartile (Q1):", Q1)

print("Second Quartile (Q2 - Median):", Q2)

print("Third Quartile (Q3):", Q3)

print("Interquartile Range (IQR):", IQR)
```

Output:

First Quartile (Q1): 18.75

Second Quartile (Q2 - Median): 27.5

Third Quartile (Q3): 36.25

Interquartile Range (IQR): 17.5

Topic 11: Visualizing Quartiles

In statistics, quartiles divide a dataset into four equal parts. They help us understand how the data is distributed and how values are spread.

When the data is arranged in ascending order:

- Q1 (First Quartile) 25% of data lies below this value
- Q2 (Second Quartile) 50% of data lies below this value (Median)
- Q3 (Third Quartile) 75% of data lies below this value

Quartiles help us analyze the central distribution of data and detect variation.

Visualizing quartiles makes it easier to:

- Understand the spread of data
- Identify the median clearly
- Detect outliers
- Compare different datasets

The most common method used to visualize quartiles is the Box Plot.

A Box Plot graphically represents quartiles.

It contains:

✓ Minimum value

✓ Q1

✓ Median (Q2)

✓ Q3

✓ Maximum value

- The box represents the middle 50% of the data (between Q1 and Q3).
- The line inside the box represents the median.
- The whiskers extend to minimum and maximum values.

Interquartile Range (IQR)

- The distance between Q1 and Q3 is called the Interquartile Range (IQR).
-
- IQR helps in identifying outliers.

If any value is very far from Q1 or Q3, it is considered an outlier.

Code:

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Sample data
```

```
data = [35, 36, 37, 38, 39, 40, 41, 42, 43, 44,  
        45, 46, 47, 48, 49, 50, 52, 53, 54, 55]
```

```
# Calculate Quartiles
```

```
Q1 = np.percentile(data, 25)
```

```
Q2 = np.percentile(data, 50)
```

```
Q3 = np.percentile(data, 75)
```

```
print("First Quartile (Q1):", Q1)
```

```
print("Median (Q2):", Q2)
```

```
print("Third Quartile (Q3):", Q3)
```

```
# Create Box Plot
```

```
plt.boxplot(data)
```

```
plt.title("Box Plot for Visualizing Quartiles")
```

```
plt.ylabel("Values")
```

```
plt.show()
```

Output:

First Quartile (Q1): 39.75

Median (Q2): 44.5

Third Quartile (Q3): 49.25

